

Reproduction Protocol — cortical organoid snRNA-seq, Fig. 2

⚠ Synthetic example. The paper, DOI, accession, repository and every number below are fabricated. Nothing here describes real work, and no citation resolves. It exists to show the report format — above all the provenance tiers, which are the point of this agent.

Paper: Example, A. et al. (2024), *Journal of Example Neuroscience*. doi:10.0000/example.2024.0001

Requested by: example user · **Date:** 2026-09-08

Scope: computational · **Target:** Fig. 2 — cell-type clustering of cortical organoids

Data: GSE000000 (fabricated) · **Code:** github.com/example-lab/example-repo (fabricated)

Goal as understood:

Reproduce the clustering and cell-type assignment shown in Fig. 2b–d from the deposited processed matrices. Wet-lab derivation of the organoids is out of scope.

0. Reproducibility at a glance

	STATUS
Processed data deposited	yes — Tier 2 filtered .h5 per sample
Authors' code available	partial — clustering yes, cell-type assignment no
Environment pinned	versions in text only; no lockfile
Sample→condition mapping	GEO characteristics_ch1
Verdict	reproducible with inferences — see §4b

Provenance tiers used below: **stated** (in the paper) · **repo** (in the authors' code) · **[inferred]** (supplied by this report) · **missing** (unrecoverable). Every step carries one.

⚠ Steps 5–6 are **[inferred]**. The paper gives no mitochondrial cutoff and no clustering resolution, so cluster counts will not match Fig. 2b exactly. Expect the same major populations in different proportions, and arbitrary cluster numbering.

1. Environment

Stack: Python 3.11 · **Source:** paper text; no lockfile deposited

PACKAGE	VERSION	SOURCE
scanpy	1.10.1	Methods, "Single-nucleus analysis"
anndata	unstated	[inferred] — whatever scanpy 1.10.1 pins
harmonypy	unstated	[inferred] — paper names Harmony, not a version
leidenalg	unstated	[inferred] — required by the stated leiden call

Reference: GRCh38-2020-A · **Annotation:** GENCODE v32 — both **stated** (Methods)

Recommended resources: 8 cores · 64 GB RAM · 40 GB disk · ~2 h wall-clock

Setup:

```
python3 -m venv .venv && ./venv/bin/pip install "scanpy==1.10.1" harmonypy leidenalg
```

⚠ Three of four package versions are inferred. Pin them yourself before treating any numerical result as a match — an unpinned scanpy minor bump has changed default HVG behaviour before.

2. Protocol

2b. Data acquisition (ETL)

Accession: GSE000000 · **Tier taken:** 2 — 10x filtered .h5

Download: 6 files · 4.2 GB · ~8 min

GSM	TITLE (AS DEPOSITED)	CONDITION	N	FILE
GSM0000001	CTRL_rep1	control	1	GSM0000001_CTRL1_filtered.h5
GSM0000002	CTRL_rep2	control	1	GSM0000002_CTRL2_filtered.h5
GSM0000003	CTRL_rep3	control	1	GSM0000003_CTRL3_filtered.h5
GSM0000004	MUT_rep1	mutant	1	GSM0000004_MUT1_filtered.h5
GSM0000005	MUT_rep2	mutant	1	GSM0000005_MUT2_filtered.h5
GSM0000006	MUT_rep3	mutant	1	GSM0000006_MUT3_filtered.h5

- **Files not used:** raw unfiltered matrices (Tier 4) — the filtered set is what Fig. 2 was built from, per Methods.
- **Sample mapping source:** GEO characteristics_ch1
- **n per group vs the paper:** matches the Fig. 2 legend (3 control, 3 mutant)

See output/EXAMPLE_notebook.ipynb for the executable version.

2c. QC, integration and clustering

1. **Load per-sample matrices** — **repo** (`analysis/load.py:L14-31`) - `sc.read_10x_h5()` per GSM, concatenated with `sample` in `.obs`
 2. **Filter cells by gene count** — **stated** (*Methods*, "Quality control") - Retain cells with $200 \leq \text{n_genes_by_counts} \leq 6000$ - **Checkpoint**: paper reports 41,782 nuclei retained across all samples
 3. **Filter genes** — **stated** (*Methods*, "Quality control") - Retain genes detected in ≥ 3 cells
 4. **Normalize and log-transform** — **repo** (`analysis/qc.py:L40-44`) - `normalize_total(target_sum=1e4)`, then `log1p`
 5. **Filter by mitochondrial fraction** — **[inferred]** - Cutoff used here: **10%**. The paper states only that "nuclei with high mitochondrial content were removed" and gives no threshold; the repo does not perform this step. 10% is a common default for nuclei preparations — it is this report's choice, not the authors'.
 6. **Cluster** — **[inferred]** - Leiden at **resolution 1.0**. The paper names Leiden and reports 14 clusters but never gives a resolution; the repo's clustering script is absent. Resolution 1.0 is the `scanpy` default and is unlikely to reproduce exactly 14 clusters. - **Checkpoint**: paper reports 14 clusters
 7. **Assign cell types** — **missing** - The paper says assignments were "curated manually against canonical markers" and lists no marker panel. The repo contains no assignment code. This step cannot be reproduced; §4a covers the remedy.
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3. What to expect when you run it

- **Should match closely**: nuclei retained after steps 2–3, the major excitatory/inhibitory/glia split, top markers per major population
- **Will differ**: UMAP coordinates (always), exact cluster count (step 6 is inferred), cluster numbering (arbitrary), DE p-values (version-sensitive)
- **Watch for**: step 5 failing loudly if `.var` lacks a mitochondrial prefix — the deposited matrices use `MT-`, not `mt-`

Re-running the authors' analysis on the authors' data is reproduction, not replication. A modest numerical discrepancy is not evidence of an error in either direction — a *sign flip* or a missing cluster is.

4. Documentation gaps

4a. Blocking — reproduction cannot proceed as written

- **Cell-type assignment (step 7)** — no marker panel, no assignment code. Fig. 2c/2d cannot be regenerated as published. *Remedy*: email the corresponding author for the marker table, or substitute a published reference-based classifier and report the substitution prominently.

4b. Undocumented parameters — filled by inference

STEP	MISSING	VALUE USED	BASIS
5	mitochondrial % cutoff	10%	[inferred] — field default for nuclei; paper says only "high mitochondrial content was removed"
6	Leiden resolution	1.0	[inferred] — scanpy default; paper reports 14 clusters but no resolution
1	anndata , harmonypy , leiden alg versions	latest compatible	[inferred] — paper names methods, not versions

4c. Undocumented steps — breaks in the chain

- Methods jump from normalization straight to "clusters were identified". Highly-variable-gene selection, scaling and PCA are all implied but never described, and the repo covers only the first of the three.

4d. Inaccessible assets

- None. The accession is public and the repository resolves.

What the authors documented well

Per-sample filtered .h5 files, a complete characteristics_ch1 mapping with genotype and batch, an explicit reference build and annotation version, and loading code that runs unmodified. The ETL section above is **stated** or **repo** throughout because of it — which is why the gaps concentrate in one place (clustering) rather than being spread through the whole pipeline.

5. Next steps

- Run cells 1–4, confirm nuclei retained against the Methods figure, then stop and compare before proceeding past the inferred steps.
- Email the corresponding author for the mitochondrial cutoff, clustering resolution, and marker panel — three short answers unblock §4a and §4b entirely.
- If extending to your own samples, note the GRCh38-2020-A reference and re-quantify rather than merging count matrices across builds.

Generated by the paper reproduction agent. Steps are tiered **stated**, **repo**, **[inferred]**, or **missing** — inferences are the report's, not the authors'. No data files were downloaded; the notebook ships unrun. Corrections welcome.